13 Each weekday Keira drives to work with her son Kaito. She always sets off at 8.00 a.m. She models her journey time, x minutes, by the distribution $X \sim \mathrm{N}(15, 4)$.

Over a long period of time she notes that her journey takes less than 14 minutes on 7% of the journeys, and takes more than 18 minutes on 31% of the journeys.

(i) Investigate whether Keira's model is a good fit for the data. **[3]**

Kaito believes that Keira's value for the variance is correct, but realises that the mean is not correct.

(ii) Find, correct to two significant figures, the value of the mean that Keira should use in a refined model which does fit the data. **[2]**

Keira buys a new car. After driving to work in it each day for several weeks, she randomly selects the journey times for n of these days. Her mean journey time for these n days is 16 minutes. Using the refined model she conducts a hypothesis test to see if her mean journey time has changed, and finds that the result is significant at the 5% level.

(iii) Determine the smallest possible value of n. **[5]**

13(i)	

13(ii)	
13(iii)	